Stat 301 Notes

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Course Goals for STAT 301

STAT 301 is an introduction to the theory underlying statistical inference. By the end of the course, you will:

- Understand different methods of obtaining point estimates, including maximum likelihood, method of moments, and Bayesian estimation
- Understand that some estimators are better than others, and be able to evaluate estimators
- Understand the theory underlying hypothesis tests, including errors, power, size, and level of tests
- Be able to derive hypothesis tests for simple situations from first principles, including likelihood ratio tests and most powerful tests
- Be able to derive interval estimates by inverting tests and using pivotal quantities

1 Order Statistics

During World War II, the Allied Forces analyzed the markings/serial numbers found on captured German equipment to see if they could determine the strength of the German forces. Specifically, they examined the serial numbers found on tanks. Allied spies were instructed to record the serial numbers of any German tanks they observed in the field. The Allied statisticians made an assumption: that the Germans had sequentially numbered their tanks in the order in which they were produced (this turned out to be true).

Without loss of generality, assume the serial numbers were $1, 2, 3, \ldots, N$. When a tank was observed, the Allied forces recorded the serial number and developed an procedure to estimate N. Let's suppose that when one spy returned from her mission and was de-briefed, she reported the following observed serial numbers: 5, 12, 32, 43, 49, 54.

In small groups, discuss the following:

- What is the research question we want to answer?
- What is the parameter we want to estimate?
- What statistic can we use to estimate this parameter? What is your reasoning?

We're going to need to remember a **lot** of concepts from STAT 262. We'll start with this one:

Multinomial Distribution

The multinomial distribution is characterized by:

1. n identical, independent trials
2. Each trial results in one of k outcomes
3. An outcome of type i has probability p_i, where $p_1 + p_2 + \cdots + p_k = 1$ and p_i is constant from trial to trial
4. The random variables of interest are $Y_1, Y_2, \ldots, Y_k$, where Y_i is the number of trials resulting in outcome i (Note: $Y_1 + \cdots + Y_k = n$)

pmf:

$$f_{Y_1,\ldots,Y_k}(y_1, \ldots, y_k; n, p_1, \ldots, p_k) = \begin{cases} \frac{n!}{y_1! y_2! \cdots y_k!} p_1^{y_1} p_2^{y_2} \cdots p_k^{y_k} & y_i = 0, 1, 2, \ldots, n \\ & y_1 + \cdots + y_k = n \\ & i = 1, \ldots, k \\ 0 & \text{otherwise} \end{cases}$$

Why are we thinking about the multinomial? It turns out to be useful in finding the distribution of order statistics.

1.1 What are Order Statistics?

While the sample mean and sample variance can be very useful quantities in many situations, they are not always the most appropriate statistics for a particular research question. In the German tank problem, we were interested in estimating the total number of tanks the Germans had produced. A spy observed the following serial numbers: 5, 12, 32, 43, 49, 54.

- Why would the sample mean or sample variance be inappropriate here?

- Is the sample random? Why or why not?

If we want to make inferences about a parameter, we need to know the distribution of the statistic/estimator we're using. Last semester, we derived the distributions of the sample mean and sample variance, as well as the distributions of some functions of these statistics. Now, we'll turn our attention to the distributions of **order statistics**, such as the sample maximum. You've already seen these! You worked with them on the STAT 262 final project.

Definition: Order statistics

The **order statistics** of a random sample $X_1, X_2, \ldots, X_n$ are the sample values placed in ascending order. They are denoted by:

Example: A snow storm goes through Lincoln, and drops 10 inches of snow. Consider $n = 5$ randomly selected neighborhoods in Lincoln, and let X_i be the time (in hours, from the time the snow stops falling) until neighborhood i is visited by the city snowplows. Assuming independence (realistic?), the observed times are $(x_1, x_2, x_3, x_4, x_5) = (12, 7, 72, 33, 20)$. In this case, the actual observations for each neighborhood are denoted $x_2 = 7$, $x_1 = 12$, $x_5 = 20$, $x_4 = 33$, $x_3 = 72$. However, we can also consider the "ordered" random sample:

In general, $X_{(1)} =$

$X_{(2)} =$

$\vdots$

$X_{(n)} =$

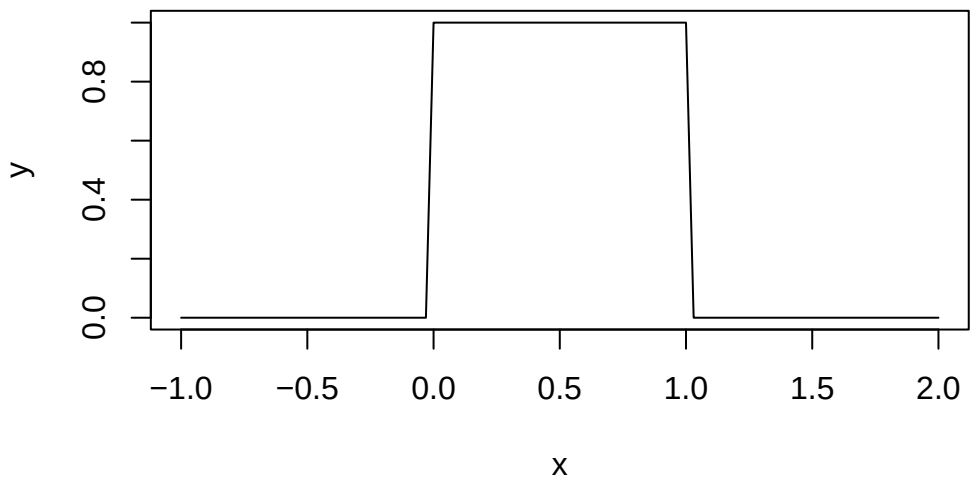
Some commonly used order statistics (and functions of order statistics) include:

However, in order to make inferences about a parameter, we need to know the distribution of the corresponding statistic/estimator. Why?

Let's consider some different distributions. Let's start with the Uniform.

1. Let $X_1, \ldots, X_n$ be a random sample from the Uniform(0,1) distribution.

```
x<-seq(-1,2,length=100)
y<-dunif(x,min=0,max=1,)
plot(x,y,type='l')
```



Let's first consider $X_{(1)}$, from samples of both $n = 2$ and $n = 100$.

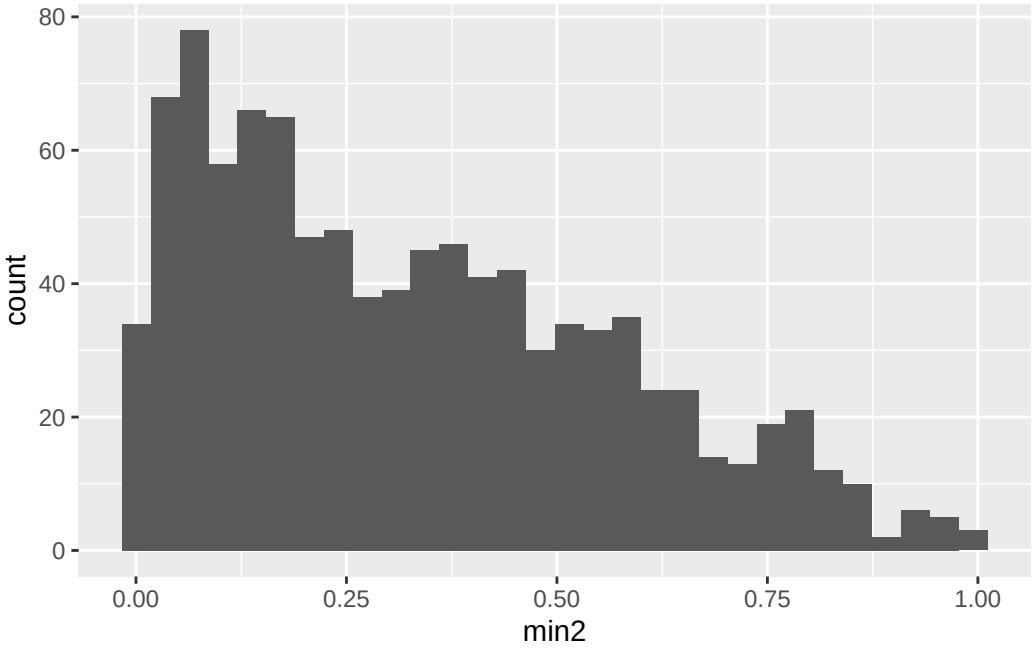
```
library(ggplot2)

unisample<-data.frame(sim=1:1000,min2=NA,min100=NA)

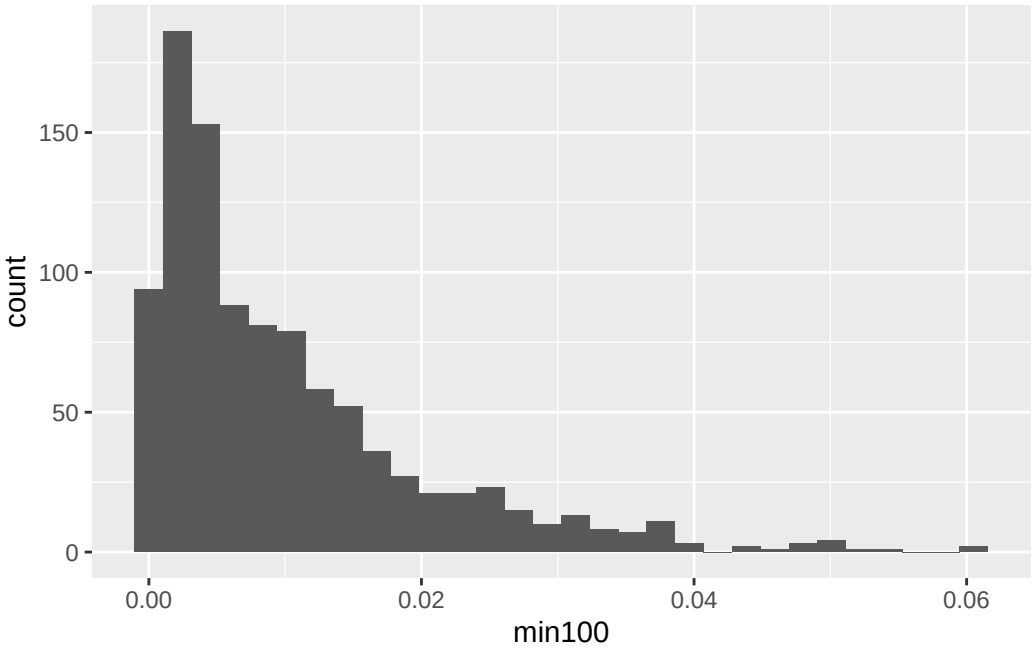
for(i in 1:1000){
  data1<-runif(2,min=0,max=1)
```

```
  data2<-runif(100,min=0,max=1)
  unisample$min2[i]<-min(data1)
  unisample$min100[i]<-min(data2)
}

ggplot(unisample, aes(min2)) + geom_histogram()
```



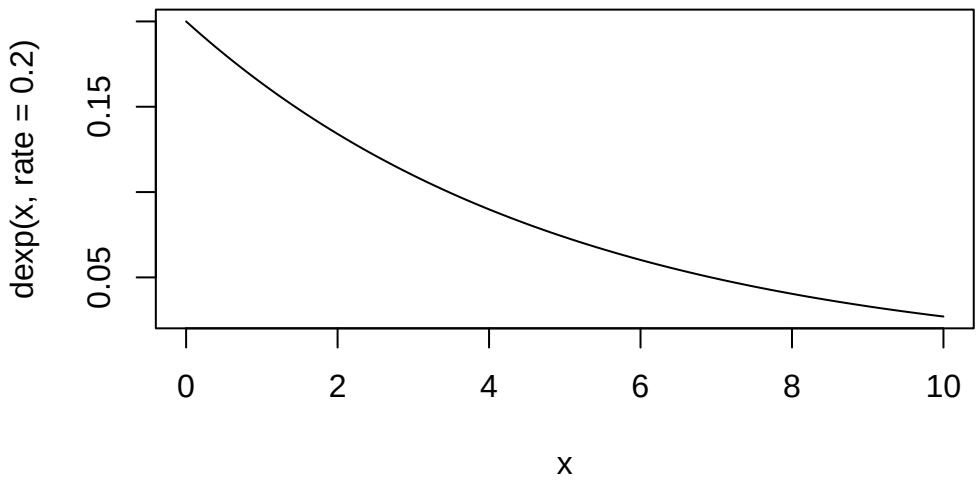

```
ggplot(unisample, aes(min100)) + geom_histogram()
```



What do you think the (empirical) distribution of the max, $X_{(n)}$ would look like?

2. Let $X_1, \ldots, X_n$ be a random sample from the Exponential($\lambda = 5$) distribution.

```
curve(dexp(x,rate=0.2),from=0, to=10)
```



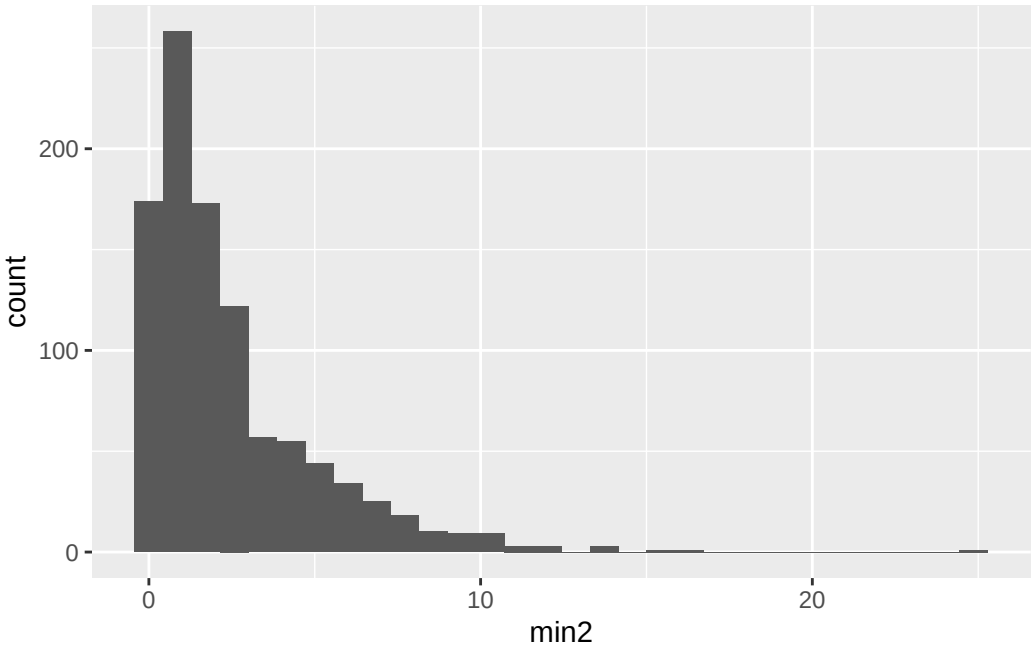
Again, let's consider $X_{(1)}$, from samples of both $n = 2$ and $n = 100$.

```
library(ggplot2)

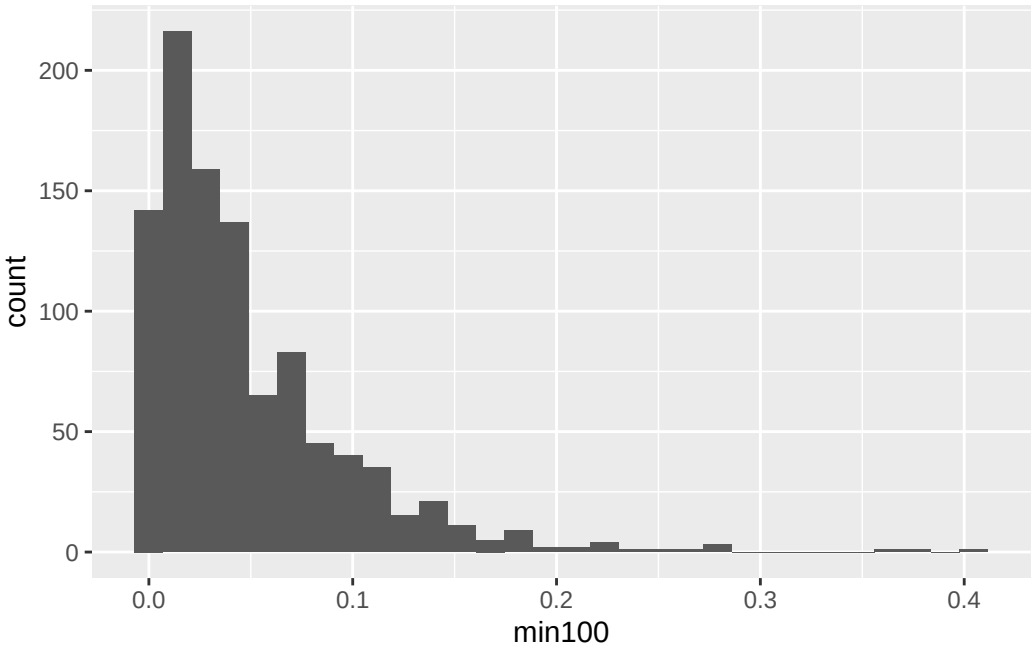
expsample<-data.frame(sim=1:1000,min2=NA,min100=NA)

for(i in 1:1000){
  data1<-rexp(2,rate=0.2)
  data2<-rexp(100,rate=0.2)
  expsample$min2[i]<-min(data1)
  expsample$min100[i]<-min(data2)
}

ggplot(expsample, aes(min2)) + geom_histogram()
```




```
ggplot(expsample, aes(min100)) + geom_histogram()
```



What about $X_{(n)}$ in this case?

What factors seems to influence what the distribution of an order statistic looks like?

1.2 The Distribution of an Order Statistic

We'll derive the theoretical distribution of a min and max through our uniform distribution example, and then we'll generalize to any distribution (and any order statistic).

Example: Let $X_1, \ldots, X_n$ be a random sample from the Uniform(0,1) distribution.

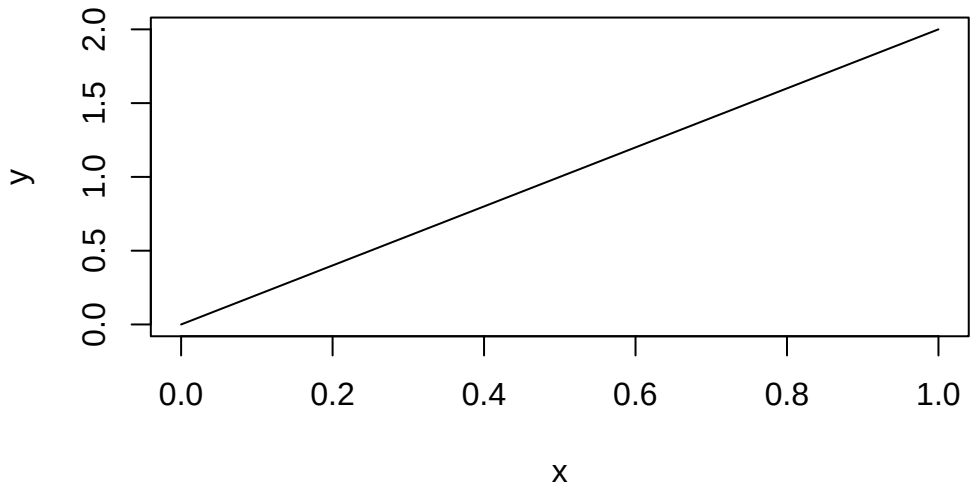
- What are the pdf and CDF of the parent population?

- Find the CDF of $X_{(n)}$

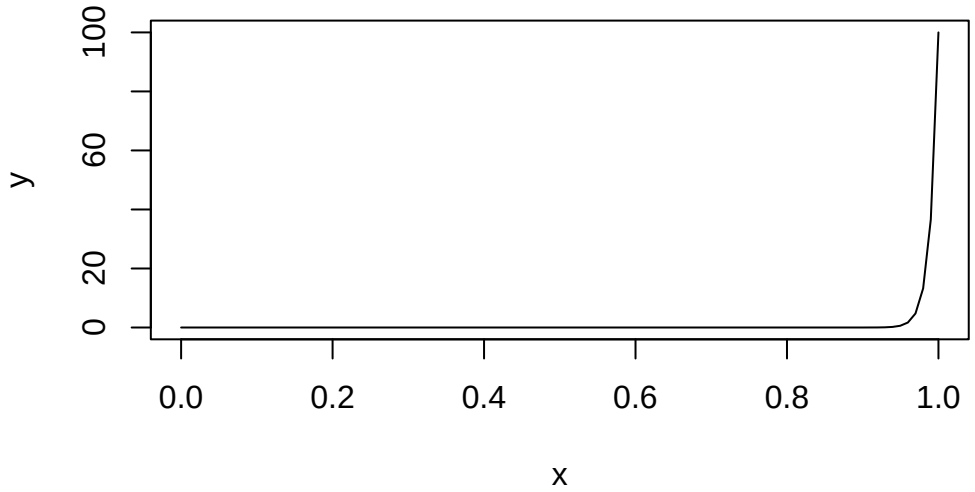
- Find the pdf of $X_{(n)}$

So, for a random sample of Uniform(0,1) random variables

- When $n = 2$, the pdf of $X_{(n)}$ is



- When $n = 100$, the pdf of $X_{(n)}$ is



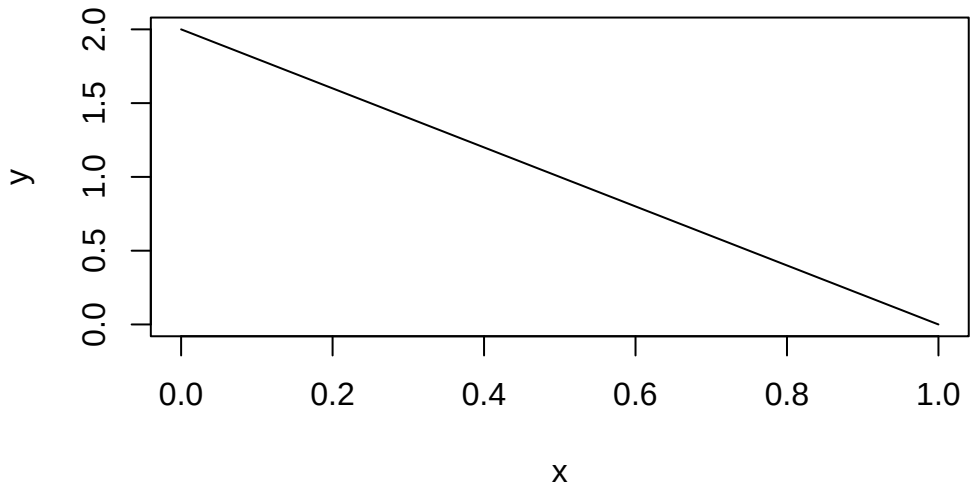
Example: Now, let's try the min.

- Find the CDF of $X_{(1)}$

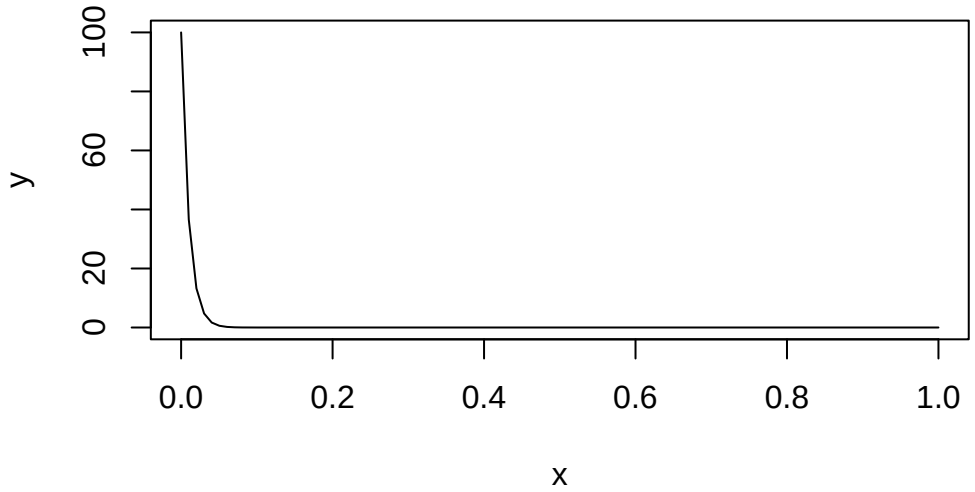
- Find the pdf of $X_{(1)}$

So, for a random sample of Uniform(0,1) random variables

- When $n = 2$, the pdf of $X_{(1)}$ is



- When $n = 100$, the pdf of $X_{(1)}$ is



What about the distributions of other order statistics, like the median? Is there a way we can cut to the pdf straightaway?

Theorem: pdf of the jth order statistic

Let $X_1, \ldots, X_n$ be independent and identically distributed *continuous* random variables with CDF $F(x)$ and pdf $f(x)$. The pdf of $X_{(j)}$ is given by

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!}[F(x)]^{j-1} f(x)[1-F(x)]^{n-j}$$

"Proof" (Please note that this is not a **real** proof) (and why haven't we used the multinomial distribution, yet?) (and are we going to have to *memorize* this????)

- For the jth order statistic, $X_{(j)}$, we have three "buckets" can toss each X_i into:
  - Group 1:
  - Group 2:
  - Group 3:
- What are the number of ways to select groups of size $j-1$, 1, and $n-j$ from a sample of size n?
- How "likely" is one of these ways?
- Putting these together:

Let's try out this theorem with the exponential distribution.

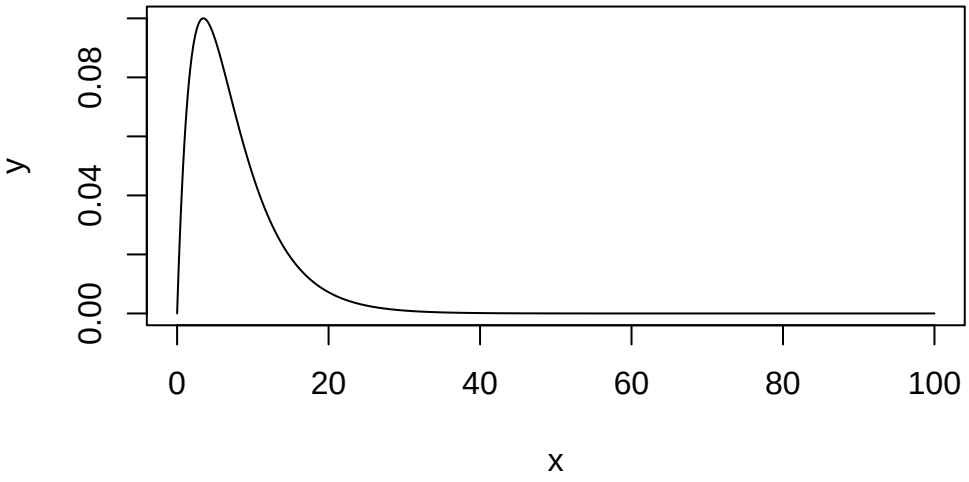
Example: Let $X_1, \ldots, X_n$ be a random sample from the exponential($\lambda = 5$) distribution.

- What are the pdf and CDF of the parent population?

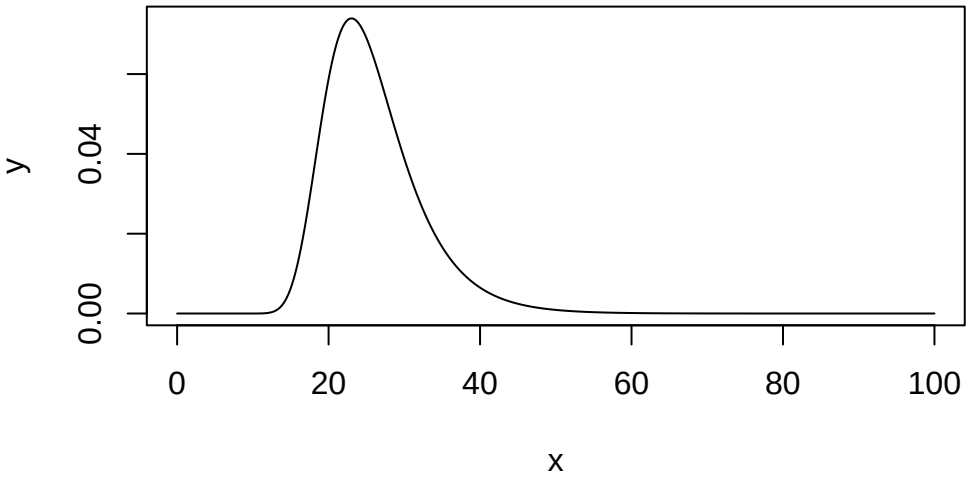
- Using the theorem, find the pdf of $X_{(n)}$

So, for a random sample of Exponential(5) random variables

- When $n = 2$, the pdf of $X_{(n)}$ is



- When $n = 100$, the pdf of $X_{(n)}$ is



We could also use this logic to find the joint distribution of any two order statistics or the joint distribution of all n order statistics.

2 Methods of Estimation

All of STAT 262, and last week, we were concerned with concepts of probability and random variables. We used them to build models of nondeterministic (i.e., random) phenomena. We focused on distributions of random variables, including function of random variables such as the sample mean. These distributions were sometimes used to determine the likelihood of a particular event or the expected value/variance of the random variable. In each of these instances, we assumed that we knew the correct probability model. Most often, however, we are interested in a numeric characteristic of a random phenomenon that cannot be computed/observed directly. For example, we might be interested in knowing

- the percent of people in Nebraska who buy organic food
- the effect of a teacher professional development program on student achievement
- the average change in blood sugar with a new diabetes drug
- the relationship between time spent playing video games and aggressive behavior
- the relationship between self-selected seating in a classroom and class performance

As we saw in classes like STAT 102 and STAT 212 (and STAT 414!), we typically do not know the true population parameter(s), like mean lifelength of patients on a particular treatment. Instead, we observe one or more random variables whose distributions depend on the parameter of interest. Then, we use statistical inference methods to analyze the observed values to gain knowledge about the unknown characteristic and/or make decisions based on the unknown characteristic. The rest of the semester we will be focusing on developing these statistical inference methods. In earlier classes we *used* these methods (under some very specific assumptions), in STAT 301 we see where these methods come from.

Definition: Statistical inference

Statistical Inference refers generally to the methods by which one makes inferences or generalizations about a population and its parameters based on statistics calculated from sample data. These methods can be classified broadly into three forms:

- **Point estimation** is the process by which an estimator of an unknown parameter, or function of parameters, is chosen and evaluated.
- **Interval estimation** is the process by which an observed value of a statistic and the statistic's probability distribution are used to create a range of plausible values of an unknown parameter, or function of parameters.
- **Hypothesis testing** is the process by which an observed value of a statistic and the statistic's probability distribution are used to evaluate the plausibility of a statement made about the unknown value of a parameter.

2.1 Point Estimation: Introduction

The basic situation is point estimation is as follows:

- We want to estimate an unknown population parameter θ (e.g., μ or λ; possibly vector-valued) or some function of parameters.

- We find an **estimator**, i.e., a function of the random variables $X_1, \ldots, X_n$: $T = W(X_1, \ldots, X_n)$ (e.g., $\bar{X}$ or S^2).

- We observe random variables $X_1, \ldots, X_n$ (not necessarily iid), where the distribution of $X_1, \ldots, X_n$ depends on the unknown parameter θ.

- We calculate an **estimate**, i.e., a function of the realized (observed) values $x_1, \ldots, x_n$: $t = W(x_1, \ldots, x_n)$ (e.g., $\bar{x}$ or s^2).

Notation: $\hat{\theta}$ or $\tilde{\theta}$ is often used in place of T to distinguish between the unknown parameter and its estimator.

Example: The Federal Aviation Administration (FAA) is interested in answering the following research question:

Is American Eagle airline more consistent when arriving in Fort Wayne, Texas than Mesa airline?

To answer this research question, the FAA randomly sampled ten *American Eagle* airline and ten *Mesa* airline flights arriving in Fort Wayne, Texas and recorded each flight's arrival delay time (in minutes). The arrival delay times for the sampled flights, in rank order, are:

AMERICAN EAGLE: -15 -13 -11 -10 -10 -4 7 37 88 142

MESA: -22 -14 -2 0 0 15 18 97 146 173

- What are the population parameters of interest?

- Identify the two random samples. What does each random variable in the random samples represent?

- Identify at least three estimators you could use to answer the research question. What would be the corresponding estimates of the parameters?

- If each group was to take its own samples (instead of the ones conveniently supplied above by the FAA), would you expect to get the same estimates? Why or why not?

- Even though we are working with the same sample data, we can get completely different estimates depending on which estimator we choose to use. Which estimator is better? How would you decide?

During our discussion of point estimation, we are going to focus on two main topics:

1. Methods of **finding estimators**, in particular Method of Moments Estimators (MMEs), Maximum Likelihood Estimators (MLEs), and Bayesian Estimators
2. Methods of **evaluating the "goodness" of statistical estimators**.

2.2 Method of Moments Estimation (MME)

The method of moments is one of the oldest methods for deriving point estimators and is due to Karl Pearson, who founded the first university statistics department at University College London. Pearson died in 1936, and during his lifetime he made many contributions to statistics, including R^2 and the χ^2 test. He was the protégé of Sir Francis Galton, cousin of Charles Darwin and the inventor of the correlation coefficient, R. In general, Pearson's method of moments generates possibly good estimators, though they can often be improved. The basic idea is that sample moments are good estimators of population moments when $X_1, \ldots, X_n$ are iid. That is, $\bar{X}$ is a good estimator of $\mathrm{E}(X)$.

Method of Moments

Let $X_1, \ldots, X_n$ be iid with common pdf (or pmf) $f(x; \underline{\theta})$ where $\underline{\theta} = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_k \end{pmatrix}$, where k is the number of parameters to be estimated.

For $j = 1, \ldots, k$, the jth **sample moment** is

and the jth **population moment** is

Because $\mathrm{E}(M_j) = \mu_j = \mu_j(\theta_1, \ldots, \theta_k)$, a **method of moments estimator (MME)** of $\underline{\theta} = (\theta_1, \cdots, \theta_k)'$ is

$$\underline{\tilde{\theta}} = \left(\tilde{\theta}_1, \cdots, \tilde{\theta}_k\right)', \quad \text{where} \quad \begin{aligned} M_1 &= \mu_1(\tilde{\theta}_1, \ldots, \tilde{\theta}_k) \\ M_2 &= \mu_2(\tilde{\theta}_1, \ldots, \tilde{\theta}_k) \\ &\vdots \\ M_k &= \mu_k(\tilde{\theta}_1, \ldots, \tilde{\theta}_k) \end{aligned}$$

Example: Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{N}(\mu, \sigma^2)$. Find a MME of $\underline{\theta} = \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}$. How do we start?

- Find the first $k = 2$ sample moments (why 2?).

- Find the first $k = 2$ population moments.

- Set the respective sample and population moments equal to each other and solve for $\underline{\tilde{\theta}}$.

Example: Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gamma}(\alpha, \beta)$. Find a MME of $\underline{\theta} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$.

Example (cont.): The lifetime in months of an electrical fuse is known to follow a gamma distribution with parameters α and β. The manufacturer wants to estimate the mean lifetime of that brand of electrical fuse. In a random sample of ten electrical fuses, the following lifetimes were observed: 3, 5, 8, 8, 4, 6, 4, 11, 6, 5. What is a method of moments estimate of the population mean?

Common Misconceptions about MMEs:

- Transformation of Data:

  The method of moments estimator (MME) is **NOT** invariant to transformation of the data. For example, if $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{N}(\mu, \sigma^2)$, but we observe $Y_1, \ldots, Y_n$ where $Y_i = e^{X_i}$, $i = 1, \ldots, n$, then a MME of $\underline{\theta} = \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}$ is not the same as the estimator we found earlier based on $X_1, \ldots, X_n$.

- Matching the First Moments:

  The first sample and population moments are **NOT** always the moments that will be matched, even if the distribution has only one unknown parameter. For example, the Laplace distribution, with pdf

  $$f_X(x; \beta) = \frac{1}{2\beta} e^{-|x|/\beta} \qquad -\infty < x < \infty, \quad \beta > 0,$$

  has $\mathrm{E}(X) = 0$. So, if $X_1, \ldots, X_n$ is a random sample from the Laplace distribution, setting the first population moment, $\mu_1 = 0$, equal to the first sample moment, $M_1 = \bar{X}$, does not yield an estimate of the unknown parameter β. However, $\mu_2 = \mathrm{E}(X^2) = 2\beta^2$, so we can use the second sample moment to get a MME.

2.3 Maximum Likelihood Estimation (MLE)

Maximum likelihood estimation is another method used to find point estimators. It was named and popularized by Sir R. A. Fisher in the early 1920s. The basic idea is that, given the sample data, choose as the estimator (e.g., $\widehat{N}$) of the parameter (e.g., N) the value that makes the observed sample result the most "likely," i.e., maximize the "likelihood."

Maximum Likelihood

Consider $\underline{X} = \mathbf{X} = (X_1, \ldots, X_n)$ which has joint pdf (or pmf) $f(\underline{x}; \underline{\theta}) = f(\mathbf{x}; \underline{\theta}) = f(x_1, \ldots, x_n; \underline{\theta})$. This joint density is considered as a function of $\mathbf{x}$ for fixed $\underline{\theta} = (\theta_1, \ldots, \theta_k)'$, where k is the number of parameters.

- The **likelihood function** (or **likelihood**) of the data is $L(\underline{\theta}|\mathbf{x}) =$

  The likelihood is viewed as a function of $\underline{\theta}$ for fixed $\mathbf{x}$.

- The **maximum likelihood estimator** (MLE) of $\underline{\theta}$, $\widehat{\underline{\theta}}_{\text{MLE}}$, maximizes the likelihood function $L(\underline{\theta}|\mathbf{x})$. That is,

  for all $\underline{\theta} \in \underline{\Theta}$ (parameter space).

- **IF** (big if) $L(\underline{\theta}|\mathbf{x})$ is differentiable and assumes a maximum on $\underline{\Theta}$, the MLE will be a solution of the equations

$$\frac{\partial}{\partial \theta_i} L(\theta|\mathbf{x}) = 0, \text{ for } i = 1, \ldots, k.$$

  If one or more solutions exist, then it should be verified which, if any, maximize $L(\underline{\theta}|\mathbf{x})$. (If a solution is a maximum, the second derivative evaluated at the solution should be < 0.)

Note: Sometimes (usually!) it is difficult to work with the likelihood function, so we work with the log-likelihood function instead. This is because any value of $\underline{\theta}$ that maximizes the likelihood, $L(\underline{\theta}|\mathbf{x})$, also maximizes the **log-likelihood**, $\ln L(\underline{\theta}|\mathbf{x})$, because the natural log function is strictly increasing on $(0, \infty)$. In other words,

So, for computational convenience, the usual form of the maximum likelihood equation is

$$\frac{\partial}{\partial \theta_i} \ln L(\theta|\mathbf{x}) = 0, \text{ for } i = 1, \ldots, k$$

.

Example: Geologists want to estimate the average magnitude of earthquakes in a region in North America. Let $X_1, \ldots, X_n$ be a random sample of earthquake magnitudes recorded for that region. Assume $X_i \sim \text{Exp}(\lambda)$, where λ is unknown. Five data points were observed: 2.2, 1.7, 3.3, 0.7 and 1.9. Find the ML estimator and estimate for λ.

Where do we start?

- Find the likelihood function, $L(\underline{\theta}|\mathbf{x})$.

IF $L(\underline{\theta}|\mathbf{x})$ is differentiable and assumes a maximum on the parameter space, then . . .

- Find the log-likelihood function, $\ln L(\underline{\theta}|\mathbf{x})$.

- Find the partial derivative of the log-likelihood function with respect to each of the parameters.

- Set the partial derivative(s) equal to zero, and solve to find $\underline{\widehat{\theta}}_{\text{MLE}}$.

- Use the data to obtain the maximum likelihood estimate of $\underline{\theta}$.

Example: A telemarketing firm is interested in estimating the probability of a sale, p. Let X_i = the number of phone calls in a day until 3 sales are made. Assume $X_1, \ldots, X_n \overset{iid}{\sim} \text{NegBin}(3, p)$, $0 \leq p \leq 1$, where p is unknown. Four data points were observed: 15, 20, 10, 22. Find $\hat{p}_{\text{MLE}}$.

- Find the likelihood function, $L(\underline{\theta}|\mathbf{x})$.

IF $L(\underline{\theta}|\mathbf{x})$ is differentiable and assumes a maximum on the parameter space, then . . .

- Find the log-likelihood function, $\ln L(\underline{\theta}|\mathbf{x})$.

- Find the partial derivative of the log-likelihood function with respect to each of the parameters.

- Set the partial derivative(s) equal to zero, and solve to find $\underline{\hat{\theta}}_{\text{MLE}}$.

- Use the data to obtain the maximum likelihood estimate of $\underline{\theta}$.

Example: Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{N}(\mu, \sigma^2)$, where both μ and σ^2 are unknown. Find the MLE of $\underline{\theta} = \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}$.

- Find the likelihood function, $L(\underline{\theta}|\mathbf{x})$.

IF $L(\underline{\theta}|\mathbf{x})$ is differentiable and assumes a maximum on the parameter space, then . . .

- Find the log-likelihood function, $\ln L(\underline{\theta}|\mathbf{x})$.

- Find the partial derivative of the log-likelihood function with respect to each of the parameters.

- Set the partial derivative(s) equal to zero, and solve to find $\underline{\widehat{\theta}}_{\mathrm{MLE}}$.

Example: Let $X_1, \ldots, X_n$ be a random sample from the distribution with pdf

$f(x;\theta) = \begin{cases} e^{-(x-\theta)} & \theta < x < \infty \\ 0 & \text{elsewhere} \end{cases}$. Find the MLE of θ.

- Find the likelihood function, $L(\underline{\theta}|\mathbf{x})$.

IF $L(\underline{\theta}|\mathbf{x})$ is differentiable and assumes a maximum on the parameter space, then . . .

So now what?!?!?!? What's going on here? Let's start over.

- Find the likelihood function, $L(\underline{\theta}|\mathbf{x})$ (REMEMBER: $L(\underline{\theta}|\mathbf{x})$ is a function of $\underline{\theta}$, and the $x_i s$ are fixed.)

- Graph the likelihood function, $L(\underline{\theta}|\mathbf{x})$.

Example: Let $X_1, \ldots, X_n$ be iid RVs with pdf $f(x;\alpha,\beta) = \begin{cases} \frac{\alpha}{\beta^\alpha} x^{\alpha-1} & 0 < x < \beta \\ 0 & \text{elsewhere} \end{cases}$, where $\alpha, \beta > 0$. Find the MLEs of α and β.

Example (cont.):

The lengths (in mm) of cuckoos' eggs found in hedge sparrow nests can be modeled with this distribution. For the data:

$$22.0, 23.9, 20.9, 23.8, 25.0, 24.0, 21.7, 23.8, 22.8, 23.1, 23.1, 23.5, 23.0, 23.0$$

find the MLEs of α and β.

Advantage of MLEs: Invariance!

Often, we know the MLE of an unknown parameter, but we're really interested in estimating a function of the unknown parameter. MLEs are invariant to transformations of parameter.

> Theorem: Invariance Property of MLEs
>
> If $\hat{\theta}$ is the MLE of θ, then for any function $\tau(\theta)$, the MLE of $\tau(\theta)$ is $\tau(\hat{\theta})$.

Example: Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{N}(\mu, \sigma^2)$. Earlier, we found $\hat{\mu}_{\text{MLE}} = \bar{X}$ and $\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$. Because MLEs are invariant, we can also find the MLEs of various functions of μ and σ^2.

1. The MLE of e^{μ} is

2. The MLE of μ^2 is

3. The MLE of σ/μ is

Example: Earlier, we looked at an example where a telemarketing firm was interested in estimating the probability of a sale, p. Recall that X_i = number of phone calls in a day until three sales are made, where $X_1, \ldots, X_n \overset{iid}{\sim} \text{NegBin}(3, p)$, $0 \leq p \leq 1$. We found $\hat{p}_{\text{MLE}} = \frac{3}{\bar{X}}$, and four data points were observed: 15, 20, 10, 22. Let's assume the firm is really interested in estimating the mean and the variance of the number of phone calls in a day before three sales are made. What are the ML estimators and estimates for the mean and variance of X?

Other considerations when calculating MLEs:

- Numerical optimization:

  For many problems we'll see in STAT 301, we use calculus to find $\hat{\theta}_{\text{MLE}}$ as an explicit function of the data, e.g., $\hat{\theta}_{\text{MLE}} = \bar{X}$. Most real life problems, however, require **numerical optimization**, which essentially involves searching the parameter space Θ for a $\hat{\theta}_{\text{MLE}}$ that maximizes the likelihood function for all $\theta \in \Theta$.

- Numerical instability:

  Because MLEs are found using a maximization process, they are susceptible to problems associated with that process, including **numerical instability**. Problems occur when the likelihood function is very flat in the neighborhood of the maximum or when there is no finite maximum.

2.4 Bayes Estimation

In both STAT 262 and so far in STAT 301 we've been dealing with iid random variables $X_1, \ldots, X_n$ with pdf/pmf $f(\mathbf{x}; \underline{\theta})$ where $\underline{\theta}$ is an unknown parameter (a fixed constant, or vector of constants). Another way to formulate the problem of estimating θ is by starting with the model above, and adding to it the assumption that θ is a random variable itself with a known distribution (called the **prior** distribution, $\pi(\theta)$). That is, we assume that $\theta \sim \pi(\theta)$, where $\pi(\theta)$ is a pdf or pmf. This means

Example: Suppose that a machine which stamps out parts for cars is examined to see what fraction θ of defectives are being produced. On any given day, 10 pieces of the machine's output are examined, with the observations denoted as $X_1, \ldots, X_n$, where $X_i = 1$ if the ith piece is defective and $X_i = 0$ if it is nondefective. These can be viewed as random sample of size 10 from the Bernoulli density:

which indicates the probability a given part is defective is equal to the unknown θ. The joint density of the 10 variables $X_1, \ldots, X_{10}$ is

There is additional information available about θ. The researcher has observed on various days the value of θ changes, and it appears the change can be represented as a random variable with the density

$$\pi(\theta) = 6\theta(1-\theta), \text{ for } 0 \leq \theta \leq 1.$$

How can this additional information be used to estimate θ_*, where θ_* is the value $\theta \in \Theta$ was equal to on the day the sample was drawn?

To make use of this additional information, we're going to need to introduce some new notation and terminology. We also need to remember a few things from STAT 262.

We now have $n+1$ random variables: $X_1, \ldots, X_n$ and θ.

- The conditional distribution of $X_1, \ldots, X_n$ given θ is
- We assume a (marginal) prior distribution for θ,
- The joint distribution of $(X_1, \ldots, X_n)$ and θ is
- The **posterior distribution** is the conditional distribution of θ given $x_1, \ldots, x_n$.

The posterior distribution contains all information about θ contained in $(x_1, \ldots, x_n)$

Bayesian Point Estimators

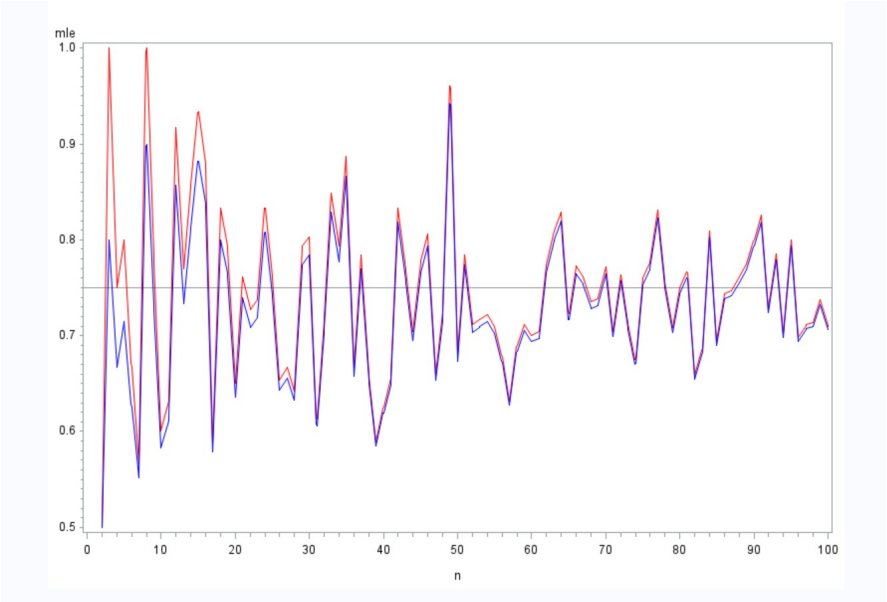
Estimators of θ are derived from the posterior distribution $\pi(\theta|x_1, \dots, x_n)$. For example,

1. $\mathrm{E}(\theta|x_1, \dots, x_n)$
2. Median of $\pi(\theta|x_1, \dots, x_n)$
3. Mode of $\pi(\theta|x_1, \dots, x_n)$

Example: Let $X_1, \dots, X_n$ be iid Bernoulli(p), and assume $p \sim \text{Beta}(\alpha, \beta)$.

Example: Again, let $X_1, \ldots, X_n$ be iid Bernoulli(p). This time, assume $p \sim$ Uniform(0,1).

To see how the posterior mean and the MLE ($\hat{p}$) compare as n increases, the graph depicts the MLE in red and the posterior mean in blue. The data were generated to have a true p of 0.75.



Bayesian analysis and estimation are based on a fundamentally different philosphy than likelihood/frequentist based methods. Why use Bayesian approaches?

- Allow the user to account for uncertainty in the systems we're trying to model, and more realistically capture the data-generating process
- Allow the user to build custom models, rather than try to shove a complex problem in an existing box
- More intuitive interpretations for some quantities (we'll see this with interval estimates)
- Some models can only be fit using Bayesian methods

2.5 What's Next?

We have seen several methods of obtaining point estimators. This leads to question: when faced with the choice of two or more estimators for the same parameter, which do we choose? Which estimator is "better"? There is not a universally appropriate measure of "goodness," but we can define some properties that will help us decide whether one (frequentist) estimator is better than another. In this context, we'll be thinking of θ as a scale parameter.

Example: Earlier, we looked at the German tank problem, where we were interested in estimating the total number of German tanks produced during World War II. Two estimators were proposed: $X_{(n)}$ and $2\bar{X}$. What properties could we use to compare the "goodness" of each estimator? That is, what properties do we want a "good" estimator to possess?

3 Evaluating Estimators

For a particular estimation problem there may be several (maybe even many) competing estimators. We'll spend the next bit of time answering the questions:

- Are some estimators better, in some sense, than others?
- If so, in what sense?

Right off, we acknowledge that some estimators are indeed better than others. For example, if $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$ we have two competing estimators for λ: $\bar{X}$ and S^2 (why?). It turns out that $\bar{X}$ is a better estimator than is S^2. To see why, we'll start by discussing some properties of estimators that will help us decide which are better.

3.1 Criteria for Good Estimators

"Good" estimators

A good estimator is **unbiased**. An estimator $\hat{\theta}$ is an unbiased estimator of θ if:

$$\text{E}(\hat{\theta}) = \theta.$$

In addition to being unbiased, a good estimator has small variance. If $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of θ, then $\hat{\theta}_1$ is a better estimator (or, a more efficient estimator) than $\hat{\theta}_2$ if:

$$\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2).$$

Remember that estimators are statistics, and are themselves RVs. As such, they have their own distributions.

If an estimator of a parameter (or function of parameters) is unbiased and has the smallest variance of all possible unbiased estimators of that parameter (or function of parameters), then the estimator is the **best unbiased (or most efficient) estimator** for that parameter (or function of parameters).

Definition: Bias

Let $\hat{\theta}$ be an estimator for a parameter θ. The **bias** of $\hat{\theta}$ is

$$\text{Bias}(\hat{\theta}) =$$

An estimator whose bias is 0 (i.e., $\text{E}(\hat{\theta}) = \theta$) is an **unbiased estimator** for θ.

It turns out these two properties (bias and variance) can be combined into a single measure: the mean-square error.

Definition: Mean-square error

If $\hat{\theta}$ is an estimator of θ, the **mean-square error** of $\hat{\theta}$ is:

$$\text{MSE}(\hat{\theta}) =$$

The MSE has the following property:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Proof:

If an estimator is unbiased, the MSE is equal to the variance.

Example: Suppose Y_1, Y_2, Y_3 are iid Uniform$(0, \theta)$ random variables. Consider the following estimators of θ: $\widehat{\theta}_1 = Y_1$, $\widehat{\theta}_2 = 2Y_1$, $\widehat{\theta}_3 = Y_1 + Y_2$, $\widehat{\theta}_4 = Y_{(n)}$, $\widehat{\theta}_5 = \bar{Y}$.

- Which of these estimators are unbiased for θ?

- Considering only the unbiased estimators, which has smallest MSE?

3.2 "Best" Unbiased Estimators

In practice, we restrict our attention to unbiased estimators if at all possible (there are some situations where unbiased estimators do not exist), so we only need to compare variances (since MSE = Variance in this situation). However, even after we rule out biased estimators, we may still have several competing estimators (see above). To decide which is "best," we need merely to find the one with the smallest variance.

Definition: Best unbiased estimator

An estimator $\hat{\theta}^*$ is **best unbiased** for θ if

$\hat{\theta}^*$ is also called a **uniformly minimum variance unbiased estimator** (UMVUE).

Example: Suppose again that Y_1, Y_2, Y_3 are iid Uniform$(0, \theta)$ RVs.

- Based on what we did in the previous example, construct an unbiased estimator of θ based solely on $\bar{Y}$ and another based solely on $Y_{(n)}$.

- Between these 2 unbiased estimators, which has the smallest MSE? Intuitively, why does it make sense that this estimator would be "better" for estimating θ than the other? With a different distribution (the exponential, say) would this still hold true? Why or why not?

3.2.1 Data Reduction

The distributions we've discussing are **parametric** statistical models, $f(x;\theta)$. That is, the random variable X has a distribution which depends on an unknown parameter θ (possibly vector-valued).

- Typically, we collect a random sample $X_1, \ldots, X_n$, and the information in the observed values $\{x_1, \ldots, x_n\}$ is used to make inferences/decisions about θ
- The random variables $X_1, \ldots, X_n$ contain information relevant to estimating θ, but may also contain information that is irrelevant for estimating θ
- Often a **summary statistic** $T(X_1, \ldots, X_n)$ is computed from the random sample, and we hope that $T(\mathbf{X})$ will be useful in making inferences about θ
- Summary statistics are most useful if they contain all information relevant to θ, but toss out the irrelevant information
- A summary statistic is a **sufficient statistic** if a statistician who knows the value of $T(\mathbf{X})$ can do just as good a job of estimating the unknown parameter θ as a statistician who knows the observed values of the entire random sample.

3.2.1.1 Sufficiency

Definition: Sufficiency principle

A statistic $T(\mathbf{X})$ is a **sufficient statistic** for θ if the conditional distribution of the sample $\mathbf{X}$ given $T(\mathbf{X})$ does not depend on θ. If $T(\mathbf{X})$ is a sufficient statistic for θ, then any inference about θ should depend on the sample $\mathbf{X}$ only through the value $T(\mathbf{X})$.

- *Discrete Distributions*

  The statistic $T(\mathbf{X})$ is a sufficient statistic for θ if and only if

  $$P(X_1 = x_1, \dots, X_n = x_n | T(\mathbf{X}) = t) = \frac{P(X_1 = x_1, \dots, X_n = x_n, T(\mathbf{X}) = t)}{P(T(\mathbf{X}) = t)}$$

  does not depend on θ

- *Continuous Distributions*

  Let $f(\mathbf{x}; \theta)$ be the joint pdf of $(X_1, \dots, X_n)$ and let $q(T(\mathbf{X}); \theta)$ be the pdf of $T(\mathbf{X})$. The statistic $T(\mathbf{X})$ is a sufficient statistic for θ if and only if $\frac{f(\mathbf{x};\theta)}{q(T(\mathbf{X});\theta)}$ does not depend on θ.

Example: Let $X_1, \dots, X_n$ be iid Binomial$(n = 1, p)$ RVs. Consider $T(\mathbf{X}) =$

Example: Let $X_1, \dots, X_n$ be iid Uniform$(0, \theta)$ RVs. Consider $T(\mathbf{X}) = X_{(1)}$.

Although the definition of a sufficient statistic is easy to understand intuitively, it turns out to be not much help. Why?

Instead, we'll appeal to the following theorem.

Theorem: Factorization Theorem

Let $f(\mathbf{x}; \theta)$ be the joint pdf (or pmf) of $\mathbf{X} = (X_1, \dots, X_n)$. A statistic, $T(X_1, \dots, X_n) = T(\mathbf{X})$, is sufficient for θ if and only if there exist functions $g(T(\mathbf{x}); \theta)$ and $h(\mathbf{x})$ such that for all sample points $\mathbf{x} = (x_1, \dots, x_n)$ and all θ

$$f(\mathbf{x}; \theta) =$$

Example: Let $X_1, \ldots, X_n$ be iid $\mathrm{N}(\mu, \sigma^2)$ RVs, where σ^2 is known.

Example: Let $X_1, \ldots, X_n$ be iid $\mathrm{N}(\mu, \sigma^2)$ RVs, where both μ and σ^2 are unknown.

Example: Let $X_1, \ldots, X_n$ be iid $\mathrm{Uniform}(0, \theta)$ RVs.

It turns out there are **many** sufficient statistics for a parameter. If $T(\mathbf{X})$ is sufficient for θ, then any one-to-one function of $T(\mathbf{X})$ is also sufficient for θ. For example, in $\text{N}(\mu, \sigma^2)$ scenario, with σ^2 known, we found that $\bar{X}$ is sufficient for μ. This means any one-to-one function of $\bar{X}$ is also sufficient:

In fact, the entire random sample $T(\mathbf{X}) = (X_1, \dots, X_n)$ is always sufficient (which makes sense! A statistician using the entire random sample should do just as a good a job of estimating θ as another statistician using the entire random sample).

If there are lots of sufficient statistics, are we in any better shape? Have we gotten any closer to figuring out which estimator to use? To figure this out, we'll need another property of (some) statistics: completeness.

3.2.1.2 Completeness

So far, we've talked about a few desirable properties of estimators: unbiasedness and sufficiency. We're still on the hunt for a "best" unbiased estimator: a UMVUE. We talked earlier about the fact that UMVUEs have the smallest variance possible for an estimator of a particular parameter (or function of parameters). Sufficiency (and, more specifically, completeness) help us in the search for **the** minimum variance estimator. How do we know if a statistic is complete?

Definition: Complete statistic

Let $f(t;\theta)$ denote the family of pdfs/pmfs of a statistic $T(\mathbf{X}) = T(X_1, \dots, X_n)$. The family of probability distributions is called **complete** if $\text{E}[g(T)] = 0$ for all θ implies that $\text{P}(g(T) = 0) = 1$ for all θ. Equivalently, $T(\mathbf{X})$ is called a **complete statistic**.

This might be the strangest definition we'll see all semester! It's saying that, if a statistic is complete, then any unbiased estimator of 0 based on that statistic should be 0 itself. What?!?!? Why do we care about unbiased estimators of 0? Here's why …we want **the** best estimator (the UMVUE). Completeness ensures we end up with **the** UMVUE, not **a** UMVUE. Basically, if a statistic is complete, then 2 different functions of T cannot have the same expected value.

Example: Let $X_1, \dots, X_n$ be iid Uniform$(0, \theta)$ RVs.

Gross! Do we really have to go through this awfulness to find a complete statistic? Luckily, if a pdf/pmf belongs to a special family of distributions, our lives get a lot easier. We hope (hope! hope! hope!) that our pdf/pmf is an **exponential family**.

Definition: Exponential family

A family of pdfs/pmfs is called an **exponential family** if it can be written as:

$$f(x;\theta) =$$

where $h(x)$ and $t_1(x), t_2(x), \dots, t_k(x)$ are real-valued function of X alone (not of θ) and $c(\theta)$ and $\omega_1(\theta), \omega_2(\theta), \dots, \omega_k(\theta)$ are real-valued functions of θ alone (not of X).

And this is important because . . . ????

Theorem: Exponential families and completeness

Let $X_1, \dots, X_n$ be iid RVs with with a pdf/pmf that belongs to an exponential family. That is,

$$f(x;\theta) =$$

Then the statistic

$$T(\mathbf{X}) =$$

is complete and sufficient for θ (so long as the parameter space Θ contains an open set in $\mathbb{R}^k$. We're not going to worry about this.)

Example: Let $X_1, \ldots, X_n$ be iid Normal(μ, σ^2) RVs.

Example: Let $X_1, \ldots, X_n$ be iid Geometric(p) RVs.

Example: Let $X_1, \ldots, X_n$ be iid Beta(α, β) RVs.

Example: Let $X_1, \ldots, X_n$ be iid Uniform$(-\theta, 0)$ RVs.

3.3 Sufficiency, Completeness, and Best Estimation

Our goal (still!) is to use $(X_1, \ldots, X_n)$ to find the best (i.e., smallest variance) unbiased estimator for an unknown parameter θ (or function of parameters). It turns out that sufficiency and completeness help us achieve this goal.

Theorem: Lehmann-Scheffè

Let U be a complete sufficient statistic for θ and let $T(U)$ be any estimator based **only** on U. Then $T(U)$ is the unique, best unbiased estimator (UMVUE) of $\tau(\theta) = \mathrm{E}[T(U)]$.

FINALLY, we have a method for obtaining a best unbiased estimator:

COMPLETENESS + UNBIASED ESTIMATOR = UMVUE

Example: Let $X_1, \ldots, X_n$ be iid Binomial(m, p) RVs. We want the UMVUE of p.

Example: Let $X_1, \ldots, X_n$ be iid N$(\mu, 1)$ RVs. We want the UMVUE of μ^2.

Example: Let $X_1, \ldots, X_n$ be iid Geometric(p). We want the UMVUE of p.

What if we can't find an unbiased estimator based on the complete sufficient statistic?

Theorem: Rao-Blackwell + Lehmann-Scheffè

Let U be a complete sufficient statistic for θ, and let $h(X_1, \dots, X_n) = h(\mathbf{X})$ be **any** unbiased estimator of $\tau(\theta)$. Then

$$T(U) = \mathrm{E}[h(\mathbf{X})|U]$$

is the UMVUE of $\tau(\theta)$.

Example: (continued) Let $X_1, \dots, X_n$ be iid Geometric(p). We want the UMVUE of p.

Example: Let $X_1, \ldots, X_n$ be iid Binomial$(n = 1, p)$ RVs. We want the UMVUE of $p(1-p)$.

4 Hypothesis Testing

Definition: Statistical inference

Statistical Inference refers generally to the methods by which one makes inferences or generalizations about a population and its parameters based on statistics calculated from sample data. These methods can be classified broadly into three forms:

- **Point estimation** is the process by which an estimator of an unknown parameter, or function of parameters, is chosen and evaluated.
- **Interval estimation** is the process by which an observed value of a statistic and the statistic's probability distribution are used to create a range of plausible values of an unknown parameter, or function of parameters.
- **Hypothesis testing** is the process by which an observed value of a statistic and the statistic's probability distribution are used to evaluate the plausibility of a statement made about the unknown value of a parameter.

We typically want to use the information contained in sample data to make inferences/decisions about a population and/or its paramters. So far, we've talked about how to find estimators, and how to evaluate the "goodness" of these estimators. Sometimes we are interested in not only estimating the parameter, but also in evaluating the plausibility of some statement about the unknown parameter. For example, we might be interested in asking questions like:

- does a new vaccine decrease the incidence rate of the flu?
- does the addition of citric acid to water extend the life of cut roses?
- does one section of a course understand the material in a class better than another section?
- does a professional development program for teachers increase student learning?

In each case, there is an underlying parameter θ:

- incidence rate of the flu
- cut rose lifespan
- "amount" of student understanding
- "amount" of student learning

The goal of hypothesis testing is to determine if θ changes in specified ways when an element of the system is changed. For example, does the incidence rate of the flu decrease when the new vaccine is used in place of the old vaccine? Does the lifespan of a cut rose increase when citric acid is added to the water? A

hypothesis test is a rule used to choose between two complementary hypotheses. Just as we saw that not all estimators are created equally, not all hypothesis tests are equally "good." We'll spend the next few weeks discussing hypothesis tests, including new definitions and notation, choosing between tests, and two methods for deriving hypothesis tests.

4.1 Elements of a Hypothesis Test

Hypotheses

The objective of a hypothesis test is to answer a question about an unknown population parameter θ, which can take on any value in the parameter space, Θ. Remember that the parameter space gives all of the possible values of the parameter.

The parameter space Θ is partitioned into two complementary spaces: Θ_0 and Θ_a. Because this is a partition, we assume that $\Theta = \Theta_0 \cup \Theta_a$ and $\Theta_0 \cap \Theta_a = \emptyset$. These two complementary spaces are used to form the corresponding complementary **hypotheses** for the question we wish to answer:

- **Null hypothesis** (H_0): the unknown $\theta \in \Theta_0$

  The null hypothesis can be thought of as the "no change" hypothesis. It is often a statement that changing the element of the system does not result in a change in the parameter θ (though not always).

- **Alternative Hypothesis** (H_a): the unknown $\theta \in \Theta_a$

  The alternative hypothesis is the complement of the null hypothesis. This means it is often the "change" hypothesis; changing the element of the system does indeed result in a change in the parameter θ.

Example: Are more than half of all "Henry Moore" prints for sale on auction sites fake?

What are the population parameter, null and alternative hypotheses, and the parameter space for the research question? What are Θ_0 and Θ_a?

Test Function

A hypothesis test is a rule by which the data are used to choose between H_0 and H_a. This rule is an indicator function that will lead us to one of two decisions:

$$\phi(X_1, \ldots, X_n) = \begin{cases} 1 & \text{reject } H_0 \\ 0 & \text{fail to reject } H_0 \end{cases} .$$

The set $\{X_1, \ldots, X_n\}$ such that $\phi(X_1, \ldots, X_n) = 1$ is the **rejection region**.

Example: Suppose an art expert will determine the authenticity of a random sample of 15 purported Henry Moore prints.

What would be a reasonable rule for deciding between the two hypotheses? What factors are important to consider when determining the rule? What factors do you think make a test "good?"

Data

We observe random variables $X_1, \ldots, X_n$ (not necessarily iid), where the distribution of $X_1, \ldots, X_n$ depends on the unknown θ.

Example: Suppose the 15 random variables in our sample had the following observed values: 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, where a 1 means the print was fake and a 0 means the Henry Moore was authentic.

Test Statistic

We calculate the value of the **test statistic**, which is a function of the realized (observed) values $x_1, \ldots, x_n$, on which the decision will be based. The test statistic is specified by the rule $\phi(X_1, \ldots, X_n)$.

Example: What is the observed value of the test statistic specified above?

Decision

We make a decision based on the value of test statistic. We either reject H_0 or fail to reject H_0.

Example: Based on the decision rule above and the observed value of the test statistic, what decision would you make? How would you answer the research question?

Errors

Any time we construct a hypothesis test, there are two possible errors we can make:

- **Type I Error**: Reject H_0 when it is true
- **Type II Error**: Failing to reject H_0 when it is false

Example: In the Henry Moore example, 15 independent prints for sale on an auction site were evaluated by an expert to determine if the print is fake.

- What are the two hypotheses?

- What would constitute a Type I error? A Type II error?

- In this situation, which error do you think is worse?

Consider a test which **always rejects** H_0. That is, $\phi(X_1, \ldots, X_n) = 1$. What happens to the probability of committing each type of error?

Consider a test which **always fails to reject** H_0. That is, $\phi(X_1, \ldots, X_n) = 0$. What happens to the probability of committing each type of error?

4.2 Hypothesis Testing Strategy

Our overall strategy is: Minimize P(Type II error) subject to a bound on P(Type I error).

That is, we first specify the P(Type I error) we are willing to tolerate, and then (if possible) find the test that has the minimum P(Type II error). Equivalently, we find the test with the maximum sensitivity for a given level of specificity.

To do this, we'll use the **power function.**

Definition: Power function

The **power function**, $power(\theta)$, of a hypothesis test, ϕ, is the probability the test will lead to rejection of H_0 when the actual parameter value is θ. That is,

$power(\theta) =$

- If θ_a is a value of the parameter in the alternative space, i.e., $\theta_a \in \Theta_a$, then

The probability of a Type II error for a particular value of the parameter in the alternative space, $\theta_a \in \Theta_a$, is denoted by $\beta(\theta_a)$. This is the definition of power commonly used in applied statistics courses.

- If θ_0 is the only value of the parameter in the null space, i.e., $\theta_0 \in \Theta_0 = \{\theta_0\}$ where $H_0 : \theta = \theta_0$, then

This is also known as the **significance level** (i.e., the probability of a Type I error) of a hypothesis test, and is denoted by

The **power function** is a function of the parameter θ, where the value of the function for a given $\theta \in \Theta$ is $power(\theta)$. The power function is often displayed graphically, as a plot of $power(\theta)$ over the entire parameter space (all possible values of θ).

As mentioned above, we want a test that minimizes P(Type II error) subject to a bound on P(Type I error). What would the power function for a "good" test look like? Consider the hypotheses $H_0 : \theta = 5$ versus $H_a : \theta > 5$.

Example: A sample of size 1 (yeah, I know, not very realistic, but it's an easy example to start with) is drawn from a Uniform$(0, \theta)$ distribution. We're interested in testing $H_0 : \theta = 3$ versus $H_a : \theta > 3$. Consider the test function $\phi(X) = \begin{cases} 1 & X > c \\ 0 & X \leq c \end{cases}$.

- What should we choose as c if the significance level of the test should be $\alpha = 0.01$?

- Find the power function of the test $\phi(X)$. Sketch the power function.

- What is the power when $\theta = 3$? Why?

- What is the probability of making a Type II error when $\theta = 4$? When $\theta = 5$? When $\theta = 8$? How do these probabilities relate to the power function?

Example: A pharmaceutical company has developed a new pain medication they think will last longer than their current formulation (which is advertised to last θ_0 hours). They plan to administer the new medication to n patients, and ask them to report their pain relief time (in hours). Assume that pain relief time is distributed exponentially with mean θ. They wish to test whether they can market this drug as lasting longer than their current formulation.

- Explain, in words, what the parameter represents.

- In words and symbols, what are the null and alternative hypotheses? Explain why these are appropriate.

- A random sample of n patients is taken, and two hypothesis tests are proposed

$$\phi_M(X_1, \ldots, X_n) = \begin{cases} 1 & X_{(1)} > c \\ 0 & \text{elsewhere} \end{cases} \quad \text{and} \quad \phi_S(X_1, \ldots, X_n) = \begin{cases} 1 & \sum_{i=1}^n X_i > k \\ 0 & \text{elsewhere} \end{cases}$$

  Intuitively, which test do you think will be "better" (more powerful for a given α)? Why?

Let's explore which test is "better" using the power functions.

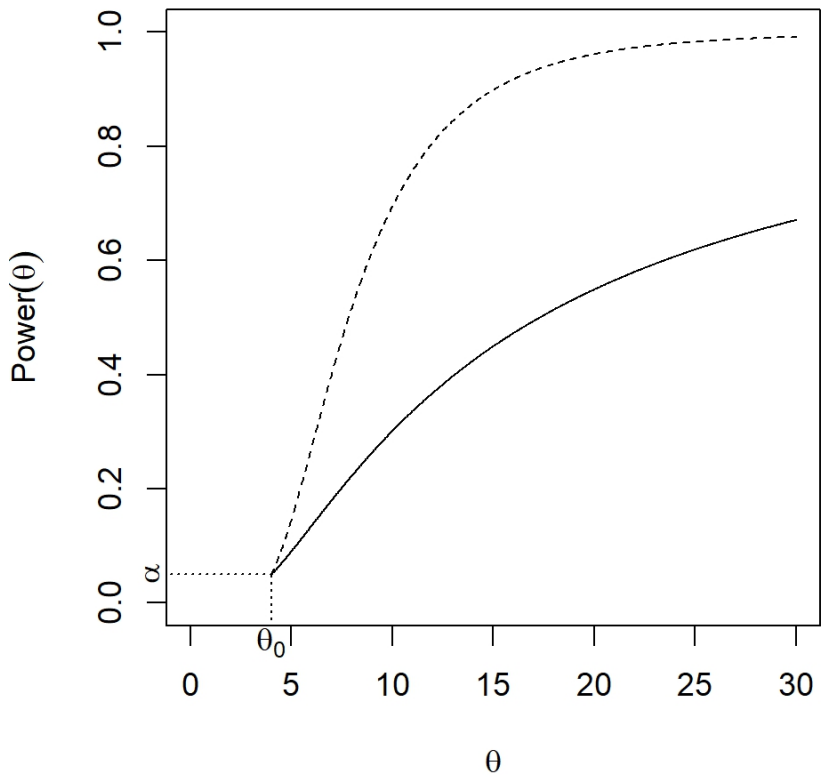
$$\phi_M(X_1, \ldots, X_n) = \begin{cases} 1 & X_{(1)} > c \\ 0 & \text{elsewhere} \end{cases}$$

$$\phi_S(X_1, \ldots, X_n) = \begin{cases} 1 & \sum_{i=1}^{n} X_i > k \\ 0 & \text{elsewhere} \end{cases}$$

Now we have the power function for each test:

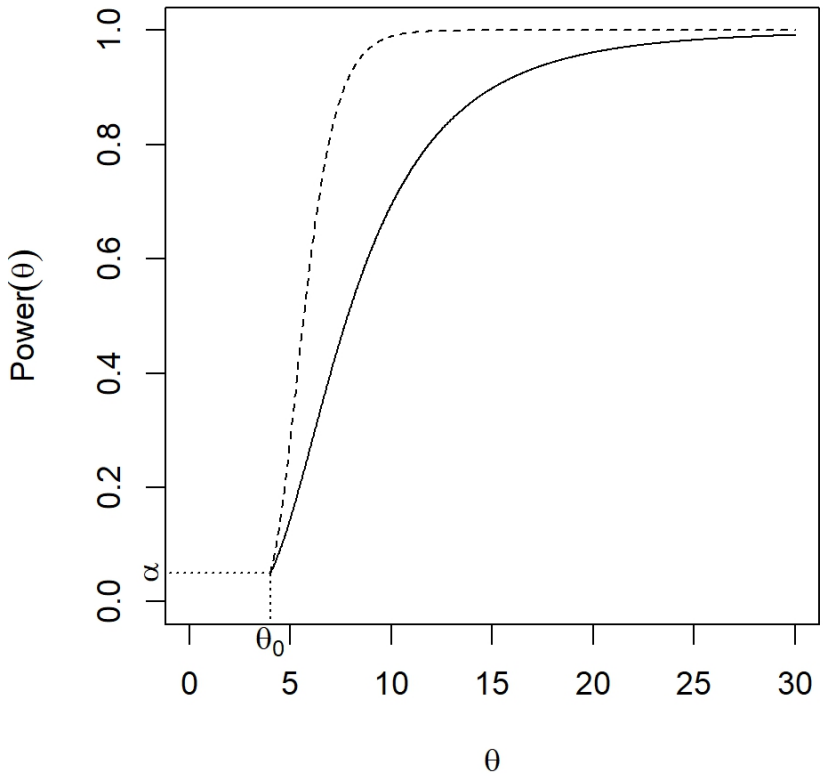
What information do we need to graph these functions?

Suppose $\theta_0 = 4$ hours. That is, the current formulation of pain medication is advertised to last, on average, 4 hours. The company wants to test whether the new drug lasts longer. They have decided to use a significance level of 0.05. The company takes a random sample of $n = 5$ patients.



If there is no clear "better" test, what do you think the power functionswill look like?

What do you think will happen to the power function of $\phi_S(X_1, \ldots, X_n)$ if the company increases the sample size from 5 to 20?



More facts about *power*(θ)

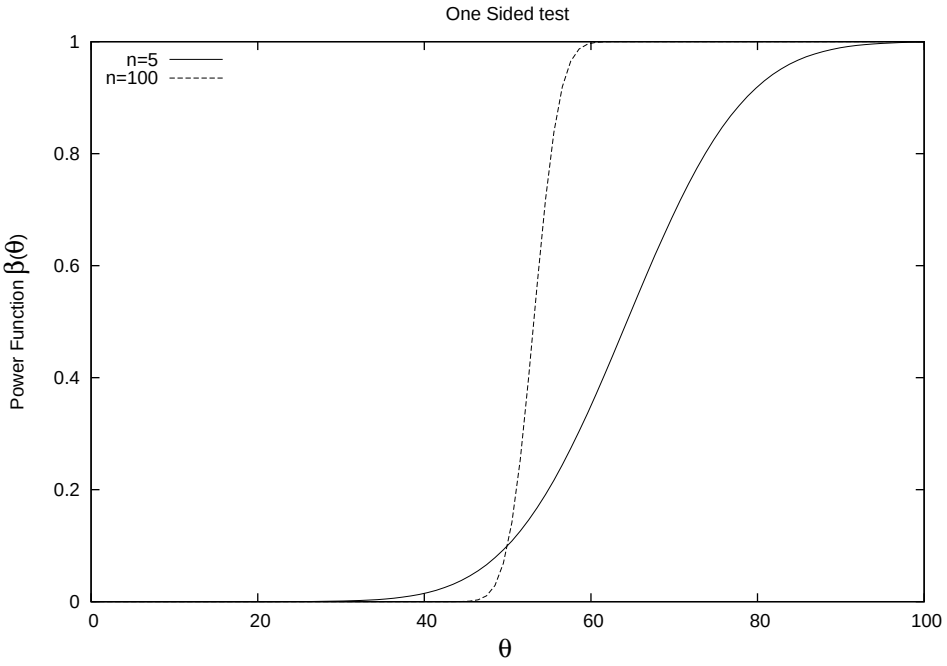
Example: Suppose $X_1, \ldots, X_n$ are iid $\mathrm{N}(\theta, \sigma^2)$, where σ^2 is known. We want to test $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta > \theta_0$. Consider the test

$$\phi(X_1, \ldots, X_n) = \begin{cases} 1 & \text{if } \frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} > c \\ 0 & \text{else} \end{cases}$$

where c is chosen to give a particular P(Type I error). Suppose we want a Type I error rate of $\alpha = 0.10$.

The power function is

Suppose $\theta_0 = 50$ and $\sigma^2 = 625$. The power function is shown for $n = 5$ and $n = 100$.



Again, note:

- The power function increases in θ.
- $power(\theta_0) = 0.10 = \Phi(1.28) = \Phi(c)$
- $power(-\infty) = 0, power(\infty) = 1$
- $power'(\theta) = \frac{d}{d\theta} power(\theta) = \phi\left(\frac{\theta-\theta_0}{\sigma/\sqrt{n}} - c\right)\frac{\sqrt{n}}{\sigma}$

This example is special in the sense that $\Theta_0 = \{\theta_0\}$. In other situations, Θ_0 will not be a single point. This is important in the following definitions.

Definition: size-α test

A test ϕ is a **size-α** test if its power function satisfies

$$\max_{\theta \in \Theta_0} power(\theta) = \alpha$$

Definition: level-α test

A test ϕ is a **level-α** test if its power function satisfies

$$\max_{\theta \in \Theta_0} power(\theta) \leq \alpha$$

A size-α test is a level-α test. For any set of null and alternative hypotheses:

$$\begin{aligned} \{\text{set of size-}\alpha\text{ tests}\} &\subset \{\text{set of level-}\alpha\text{ tests}\} \\ &\subset \{\text{set of level-}\alpha^*\text{ tests, }\alpha^* \geq \alpha\} \end{aligned}$$

Another consider is the form of the hypotheses. A hypothesis that completely specifies the distribution of the data is a **simple hypothesis**. A hypothesis that does not completely specify the distribution of the data is a **composite hypothesis**.

Example: Let $X_1, \ldots, X_n$ be iid $\mathrm{N}(\theta, \sigma^2)$, where σ^2 is known.

1. $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta > \theta_0$

2. $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta = \theta_1$

3. $\mathrm{H}_0 : \theta \leq \theta_0$ versus $\mathrm{H}_a : \theta > \theta_0$

Why does this matter?

- If H_0 is simple, then we can easily define

- If H_0 is composite, then

However, we can bound the probability of a Type I error for any $\theta \in \Theta_0$ by

$$\max_{\theta \in \Theta_0} P(\phi(X_1, \ldots, X_n) = 1) = \max_{\theta \in \Theta_0} power(\theta) = \text{ size of the test}$$

It makes sense to require

$$\max_{\theta \in \Theta_0} power(\theta) \leq \min_{\theta \in \Theta_a} power(\theta)$$

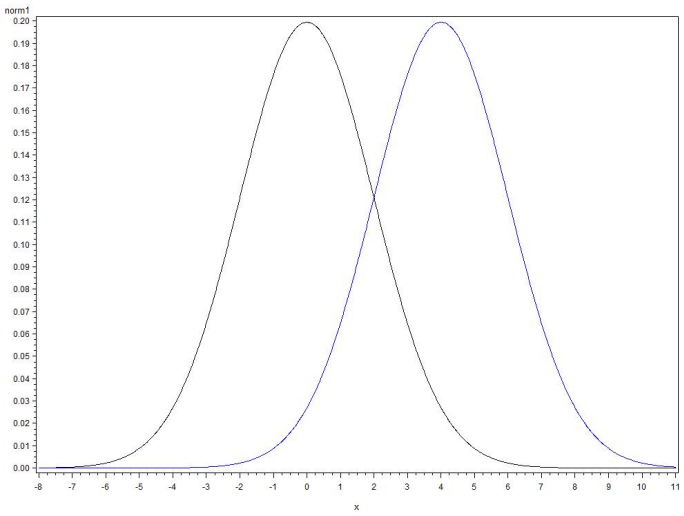
This means the power of the test is at least as large as its level of significance. That is,

A test with this property is called **unbiased**.

4.3 Likelihood Ratio Tests

Using the power function, we can determine if one test is "better" than another, but how do we find tests in the first place?

Example: Suppose we have 1 observation, and we're trying to decide between $\mathrm{H}_0 : X \sim N(0, 4)$ and $\mathrm{H}_a : X \sim N(4, 4)$.



Which hypothesis do you pick if:

- $X = 3$?
- $X = 1$?
- $X = 2$?

In each case, how did you arrive at your conclusion?

4.3.1 Constructing Likelihood Ratio Tests

We can use the "likelihood" of the two hypotheses given the data to choose between the two alternatives. Most of the time, however, we don't know θ so we'll need to estimate it. What type of estimator do you think we'll use?

Definition: Likelihood Ratio Test

Consider testing $H_0 : \theta \in \Theta_0$ versus $H_a : \theta \in \Theta_a$. The likelihood ratio test statistic is

$$\lambda(x_1, \ldots, x_n) =$$

The **likelihood ratio test** (LRT) is any test that has a rejection region of the form $\lambda(x_1, \ldots, x_n) \leq k$, where is k is chosen to achieve the desired signficance level, with $0 \leq k \leq 1$. That is, the LRT has the form $\phi(X_1, \ldots, X_n) = \begin{cases} 1 & \lambda(x_1, \ldots, x_n) \leq k \\ 0 & \text{elsewhere} \end{cases}$.

What's in this ratio?

- Numerator:

- Denominator:

And comparing them does what, exactly?

- Why must $\lambda(X_1, \ldots, X_n) \leq 1$?

- What would a value of $\lambda(X_1, \ldots, X_n)$ close to 1 indicate?

- What would a value of $\lambda(X_1, \ldots, X_n)$ close to 0 indicate?

Finding LRTs

1. Specify hypotheses: $\mathrm{H}_0 : \theta \in \Theta_0$ versus $\mathrm{H}_a : \theta \in \Theta_a$
2. Specify the likelihood function:

$$L(\theta|x_1, \ldots, x_n) = L(\theta|\mathbf{x}) = f(\mathbf{x}; \theta) = f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

3. Find $\max_{\theta \in \Theta_0} L(\theta|\mathbf{x}) = L(\hat{\theta}_0|\mathbf{x})$
4. Find $\max_{\theta \in \Theta} L(\theta|\mathbf{x}) = L(\hat{\theta}_{\text{MLE}}|\mathbf{x})$
5. Calculate $\lambda(\mathbf{x}) = \frac{L(\hat{\theta}_0|\mathbf{x})}{L(\hat{\theta}_{\text{MLE}}|\mathbf{x})}$
6. Find a critical value for $\lambda(\mathbf{x})$ to ensure a signficance level of α
7. Carry out the test and report the conclusion **in context**

Example: The diameters of table legs produced by a machine are RVs distributed as $N(\mu, 0.01)$. In order to correctly fit the table into which they will be inserted, the legs should have a mean diameter of 2.5 inches. The operator running the machine suspects that the machine is malfunctioning.

- What is(are) the population(s) of interest?

- What are the hypotheses we wish to test?

- Derive a likelihood ratio test for testing these hypotheses.

- The machine operator selects a random sample of $n = 16$ table legs, and observes a sample mean of $\bar{x} = 2.48$ inches. Does the machine need to fixed? Use $\alpha = 0.1$.

Example: Let the random variable X denote the lifetime of cut roses delivered by a florist (in days). Suppose X follows the distribution with pdf and CDF

$$f_X(x) = \begin{cases} e^{-(x-\delta)} & x \geq \delta \\ 0 & \text{elsewhere} \end{cases} \quad \text{and} \quad F_X(x) = \begin{cases} 1 - e^{-(x-\delta)} & x \geq \delta \\ 0 & \text{elsewhere} \end{cases},$$

respectively. The parameter δ depends on the amount of time between when the roses were harvested and when they were delivered. Based on historical data, you believe that, for a particular variety of rose, $\delta = 5$. You've heard an old story that adding 7-UP to the water will make the roses last longer, and want to test this theory. That is, you want to see if adding 7-UP to the water will increase the value of δ. Construct a LRT to investigate this hypothesis.

Example: During the first year of its existence, the roundabout at 14th and Superior averaged about 2 crashes per week. To help change the number of accidents, the city put up barriers to remove one lane of traffic (3 lanes were reduced to 2). After those barriers were installed, there were a total of 3 accidents in the first 5 weeks of operation. Does it appear that the barriers worked to change the number of accidents? Use $\alpha = 0.05$.

- What is(are) the population(s) of interest?

- What are the hypotheses we wish to test?

- Derive a LRT for testing these hypotheses.

4.3.2 Approximate Distribution Theory for LRTs

Under some "regularity" conditions on the distribution of the data, the large sample distribution of the LRT statistic is χ^2.

Theorem

If $X_1, \ldots, X_n$ are iid with pdf/pmf $f(x;\theta)$ and $\lambda(x_1, \ldots, x_n)$ is the likelihood ratio test statistic for testing $\mathrm{H}_0 : \theta \in \Theta_0$ versus $\mathrm{H}_a : \theta \in \Theta_a$, then under regularity conditions on $f(x;\theta)$ under H_0, as $n \to \infty$:

$$-2\ln[\lambda(\mathbf{x})] \xrightarrow{D} \chi^2_k$$

where \$k = \$[# free parameters for $\theta \in \Theta$] - [# free parameters for $\theta \in \Theta_0$]. This yields the test with (approximate) significance level α:

$$\phi(X_1, \ldots, X_n) = \begin{cases} 1 & -2\ln[\lambda(\mathbf{x})] > \chi^2_{k,\alpha} \\ 0 & \text{elsewhere} \end{cases}$$

Example (con't):

Example (Homework 5): A random sample of 100 Lincoln residents produced a total of 14 who favor allowing non-Lincoln residents to vote in Lincoln elections. An independent random sample of 100 non-Lincoln residents produced a total of 46 who favor allowing non-Lincoln residents to vote in Lincoln elections. Assume p_L is the true proportion of Lincoln residents who favor allowing non-residents to vote and p_{NL} is the true underlying proportion of non-Lincoln residents who favor allowing allowing non-residents to vote. Derive a LRT to determine if there is a significant difference between p_L and p_{NL}.

4.4 Best Tests

With estimation, we devised a method for constructing "best" estimators. With tests, what makes one test "better" than another? Can we devise a method for identifying and deriving "best" tests?

We'll start with simple versus simple hypotheses. We'll consider the task of specifying a rejection region such that it has the probability α when H_0 is true (Type I error) and the largest possible probability (i.e., highest power) when H_a is true.

Let's think about rejection regions R having the form

$$\left\{(x_1, \ldots, x_n) : \frac{f(x_1, \ldots, x_n|\theta_1)}{f(x_1, \ldots, x_n|\theta_0)} > k\right\}$$

Why does this provide a good rejection region? Let's look at a simple example.

Example: Suppose $X_1, \ldots, X_5 \sim$ Binomial(1,p). We want to test $H_0 : p = 0.2$ versus $H_a : p = 0.5$ with $\alpha = 0.35$. We'll base our test on $\sum_{i=1}^{5} X_i \sim$ Binomial$(5, p)$.

	$x = 0$	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$
$P(\sum X_i = x \mid p = 0.2)$	0.3277	0.4096	0.2048	0.0512	0.0064	0.0003
$P(\sum X_i = x \mid p = 0.5)$	0.0313	0.1563	0.3125	0.3125	0.1562	0.0312

Let's consider 2 options for the rejection region.

When we consider the power-to-level ratio:

	$x=0$	$x=1$	$x=2$	$x=3$	$x=4$	$x=5$
$P(\sum X_i = x \vert p = 0.2)$	0.3277	0.4096	0.2048	0.0512	0.0064	0.0003
$P(\sum X_i = x \vert p = 0.5)$	0.0313	0.1563	0.3125	0.3125	0.1562	0.0312
$\frac{P(\sum X_i = x \vert p=0.5)}{P(\sum X_i = x \vert p=0.2)}$	0.0955	0.3812	1.5259	6.1035	24.406	104

4.4.1 The Neyman-Pearson Lemma

This idea of constructing the rejection region by including points that contribute the most power per level is the motivation for the **Neyman-Pearson Lemma**.

Theorem: Neyman-Pearson Lemma

Consider testing $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta = \theta_1$, where $(X_1, \ldots, X_n) \sim f(x_1, \ldots, x_n|\theta)$. For a constant $k > 0$, let

$$\phi(X_1, \ldots, X_n) = \begin{cases} 1 & f(x_1, \ldots, x_n|\theta_1) > kf(x_1, \ldots, x_n|\theta_0) \\ 0 & f(x_1, \ldots, x_n|\theta_1) < kf(x_1, \ldots, x_n|\theta_0) \end{cases}$$

and let α be defined as

$$\alpha = P(\phi(X_1, \ldots, X_n) = 1|\theta = \theta_0).$$

Then, $\phi(X_1, \ldots, X_n)$ is most powerful for testing H_0 versus H_a at level α.

Example: Let $X_1, \ldots, X_n$ be a random sample from the exponential(θ) distribution. We want to test $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta = \theta_1$ where $\theta_1 > \theta_0$.

Example: Let $X_1, \ldots, X_n$ be Normal(μ, σ^2) random variables where σ^2 is known. We want to test $\mathrm{H}_0 : \mu = \mu_0$ versus $\mathrm{H}_a : \mu = \mu_1$ where $\mu_1 > \mu_0$.

The test we just derived does not depend on μ_1 The **same** test is most powerful for every $\mu_1 > \mu_0$ (though the power will differ for different values of μ_1). If a test has this property it is called **uniformly most powerful** (UMP).

> Definition: Uniformly Most Powerful (UMP) Tests
>
> A test is a **uniformly most powerful** (UMP) test of size α if it has size α and if for all parameter values in the alternative spae it has maximum power relative to all rejection regions of size α.

In practice, the Neyman-Pearson Lemma may be used to find UMP tests for composite alternative hypotheses by first finding the NP test for a particular simple alternative, and then showing the test does not depend on the specified alternative value.

Example: Let $X_1, \ldots, X_n$ be Normal(μ, σ^2) random variables where σ^2 is known. We want to test $\mathrm{H}_0 : \mu = \mu_0$ versus $\mathrm{H}_a : \mu = \mu_1$ where $\mu_1 > \mu_0$.

Like MLEs (and LRTs), the NP Lemma guarantees that most powerful tests (if they exist) will be functions of sufficient statistics. The NP Lemma says the test

$$\phi(X_1, \ldots, X_n) = \begin{cases} 1 & f(x_1, \ldots, x_n|\theta_1) > kf(x_1, \ldots, x_n|\theta_0) \\ 0 & f(x_1, \ldots, x_n|\theta_1) < kf(x_1, \ldots, x_n|\theta_0) \end{cases}$$

is most powerful.

4.4.2 Extending the Neyman-Pearson Lemma

We've already seen that the Neyman-Pearson Lemma may be used to find UMP tests for composite alternative hypotheses by first finding the most powerful test for a particular alternative and then showing that it does not depend on the specific alternative value. What about composite nulls?

Theorem: Extending the Neyman-Pearson Lemma

Let $\phi(X_1, \ldots, X_n)$ be a size-α NP test for $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta = \theta_1$ $(\theta_1 > \theta_0)$ where $\mathrm{E}(\phi(X_1, \ldots, X_n|\theta_0) = \alpha$. Suppose the test does not depend on θ_1.

- 1st extension: The NP test is uniformly most powerful for testing $\mathrm{H}_0 : \theta = \theta_0$ versus $\mathrm{H}_a : \theta > \theta_0$ (its size is still α).
- 2nd extension: If in addition the power function of NP best test is non-decreasing for all θ, then the test is UMP size-α for $\mathrm{H}_0 : \theta \leq \theta_0$ versus $\mathrm{H}_a : \theta > \theta_0$.

Example: Let $X_1, \ldots, X_n$ be Normal(μ, σ^2) random variables where σ^2 is known.

Example: Let $X_1, \ldots, X_n$ be a random sample from pdf $f(x|\lambda) = \lambda e^{-\lambda x}$ for $x > 0$. We want to test $\mathrm{H}_0 : \lambda \leq \lambda_0$ versus $\mathrm{H}_a : \lambda > \lambda_0$.

What if we want to test $H_0 : \theta \geq \theta_0$ versus $H_a : \theta < \theta_0$? What needs to change in the extensions?

Theorem: Extending the Neyman-Pearson Lemma (Less Than)

Let $\phi(X_1, \ldots, X_n)$ be a size-α NP test for $H_0 : \theta = \theta_0$ versus $H_a : \theta = \theta_1$ (θ_1 θ_0) where $E(\phi(X_1, \ldots, X_n|\theta_0) = \alpha$. Suppose the test does not depend on θ_1.

- 1st extension: The NP test is uniformly most powerful for testing $H_0 : \theta = \theta_0$ versus $H_a : \theta < \theta_0$ (its size is still α).
- 2nd extension: If in addition the power function of NP best test is _____ for all θ, then the test is UMP size-α for $H_0 : \theta \geq \theta_0$ versus $H_a : \theta < \theta_0$.

If the NP Lemma and its extensions give us a method for finding most powerful tests, why would we ever do anything different? Why even bother with LRTs and others? **UMP tests don't always exist.**

Example: Let $X_1, \ldots, X_n$ be Normal(μ, σ^2) random variables where σ^2 is known.

4.4.3 Monotone Likelihood Ratios

The monotone likelihood ratio is a condition that, when it holds, lets us find UMP tests for 1-sided alternatives without going through the NP Lemma + Extensions process.

Definition: Monotone Likelihood Ratio

A joint pdf/pmf $f(x_1, \dots, x_n|\theta)$ has a **monotone likelihood ratio** (MLR) in the statistic $T = t(X_1, \dots, X_n)$ if, for any two values of the parameter $\theta_2 > \theta_1$, the ratio $f(x_1, \dots, x_n|\theta_2)/f(x_1, \dots, x_n|\theta_1)$ depends on $(x_1, \dots, x_n)$ only through the function $t(x_1, \dots, x_n)$ and this ratio is a non-decreasing function of $t(x_1, \dots, x_n)$.

Note: if the ratio is non-increasing in $t(x_1, \dots, x_n)$ it can be made non-decreasing.

Example: Let $X_1, \dots, X_n$ be iid Poisson(λ) random variables.

Theorem: Karlin-Rubin (Greater than)

Consider testing $\mathrm{H}_0 : \theta \leq \theta_0$ versus $\mathrm{H}_a : \theta > \theta_1$. Suppose that T is a sufficient statistic for θ and the family of pdfs/pmfs of T has an MLR. Then, for any t_0, the test that rejects H_0 if and only if $T > t_0$ is a UMP level-α test, where $\alpha = P(T > t_0|\theta_0)$.

Example: Let $X_1, \ldots, X_n$ be iid Poisson(λ) random variables. We want to test $\mathrm{H}_0 : \lambda \leq \lambda_0$ versus $\mathrm{H}_a : \lambda > \lambda_0$.

Example: Let $X_1, \ldots, X_n$ be a random sample from pdf $f(x|\lambda) = \lambda e^{-\lambda x}$ for $x > 0$. We want to test $\mathrm{H}_0 : \lambda \leq \lambda_0$ versus $\mathrm{H}_a : \lambda > \lambda_0$.

MLRs are another reason to love exponential families.

Theorem: MLRs and Exponential Families

If $(X_1, \ldots, X_n)$ has a distribution in the one-parameter exponential family, i.e. has pdf

$$f(x|\theta) = c(\theta)h(x)\exp\{w(\theta)t(x)\}$$

where $w(\cdot)$ is a non-decreasing function, then the distribution of $(X_1, \ldots, X_n)$ has a MLR in $T(X_1, \ldots, X_n) = \sum_{i=1}^{n} t(x_i)$.

Example: Let $X_1, \ldots, X_n$ be a random sample from pdf $f(x|\lambda) = \lambda e^{-\lambda x}$ for $x > 0$.

Example: Let $X_1, \ldots, X_n$ be a random sample from the beta distribution with α unknown and β known.

The problem of testing $\mathrm{H}_0 : \theta \geq \theta_0$ versus $\mathrm{H}_a : \theta < \theta_0$ can also be handled by the MLR approach, but the inequalities in the Karlin-Rubin Theorem must be reversed.

Theorem: Karlin-Rubin (Less than)

Consider testing $\mathrm{H}_0 : \theta \geq \theta_0$ versus $\mathrm{H}_a : \theta < \theta_1$. Suppose that T is a sufficient statistic for θ and the family of pdfs/pmfs of T has an MLR. Then, for any t_0, the test that rejects H_0 if and only if $T < t_0$ is a UMP level-α test, where $\alpha = P(T < t_0|\theta_0)$.

4.5 P-values

So far, we've been dealing with tests with a pre-specified significance level. That is, our tests have the form $\phi(X_1, \dots, X_n) = \begin{cases} 1 & T(X_1, \dots, X_n) > k_\alpha \\ 0 & T(X_1, \dots, X_n) \leq k_\alpha \end{cases}$ where $\alpha = P(T > k_\alpha | \text{H}_0 \text{ true})$. The numerical value of α is chosen arbitrarily (or by tradition), and different researchers may disagree about what value of α should be used. This can lead to an uncomfortable situation: two researchers come to different conclusions, even with the same set of data. However, if the researchers use the same test statistic, we can overcome this problem by reporting the results in terms of the **p-value** of the test.

> Definition: p-value
>
> If $W(X_1, \dots, X_n)$ is a test statistic, the **p-value**, or **attained significance level**, is the smallest level of significance α for which the observed data indicate that the null hypothesis should be rejected. The smaller the p-value, the more compelling is the evidence that the null hypothesis should be rejected.

Example: Let's go back to the 14th and Superior roundabout example. In that case, our LRT was

$$\phi_\alpha(X_1, \dots, X_n) = \begin{cases} 1 & W(X_1, \dots, X_n) > k_\alpha \\ 0 & W(X_1, \dots, X_n) < k_\alpha \end{cases} .$$

Previously, we set $\alpha = 0.05$. We could have chosen many different signficance levels:

Notice that as α decreases, the critical value k_α increases, making it more difficult to reject the null hypothesis. This makes sense, because we are reducing our chance of making a Type I error. With the data observed, our test statistic was 6.776, and at $\alpha = 0.05$, we rejected the null hypothesis. However, if we had used a different α value, we would have reached a different conclusion.

To avoid this issue, p-values are often reported so that researchers (or, maybe more importantly, consumers of the research) can reach their own conclusions about the significance of the test results. The p-value of the test is the smallest value of α for which we would reject H_0. That is, it is the smallest value of α for

which $\phi(X_1, \ldots, X_n) = 1$. For example, if our p-value=0.024, then 0.024 is the smallest value α could be, and we would still reject H_0. For a test with significance level α, we could re-write the test rule as

$$\phi_\alpha(X_1, \ldots, X_n) = \begin{cases} 1 & \text{p-value} < \alpha \\ 0 & \text{p-value} > \alpha \end{cases}$$

P-values are convenient because the end user of the test doesn't need to look up critical values in a table. Note that smaller p-values indicate greater evidence against H_0 and larger p-values indicate lesser evidence against H_0.

There are two main ways to calculate p-values:

1. Reference distribution of the test statistic, T.

2. Simulation